

An Investigation on Location in the Round of 64 in March Madness

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1 Introduction

Every year, analysts, media personnel, and the fans of NCAA Men's College Basketball aim to predict the first round of the NCAA Men's Collegiate Basketball Championship, AKA March Madness. People consider many factors when predicting these results, one of which is location. If a team is closer to the location of their game, it seems logical that they might have a better chance of winning the game. This especially pertains to "upset picks," where a higher-seeded team, touted as underdogs, are chosen to be watched and potentially bet on because they are closer in location than their lower-seeded counterpart. This investigation takes a deep dive into March Madness data from 2011-2019 and 2022-2026 to see: **Does location give you an edge over the spread when betting?**

2 Key Terms to Know

To summarize the key terms:

- **Spread** - The amount of points the favorite is expected to win by.
 - Example 1: A spread of (-3.5) means a team is favored by 3.5 points. If they win by more than 3.5, the team "covers" and is considered to be performing above expectations. If they win by less than 3.5 or lose outright, they "don't cover" and are considered to be performing below expectations.
 - Example 2: A spread of (+6) means a team is an underdog by 6 points. If they lose by less than 6 or win outright, the team "covers" and is considered to be performing above expectations. If they lose by exactly 6 points, it is considered a "push" and is considered to be performing at expectations. If they lose by more than 6 points, they "don't cover" and are considered to be performing below expectations.

- **Distance** - The straight-line geographical distance between the location of the game and the college of the team participating in the match (in miles).
- **Seed** - The predetermined “ranking” of each team by a committee of experts in a given March Madness tournament. Out of the 64 teams, there are 4 groups of 16 teams each seeded 1-16, with the 1 seed being the best team in their respective group, and the 16 seed being the worst. Matchups in the Round of 64 will always be a 1 v 16, 2 v 15, 3 v 14, ... 8 v 9.
- **First Four** - The first 4 games played in the tournament are to whittle the initial 68 team field to 64. These are usually the matchups to determine the 11 or 16 seeds. They are NOT apart of the Round of 64 but are apart of the tournament and are typically played a few days prior to the Thursday the Round of 64 starts.
- **Round of 64** - This is the first official round of March Madness. After the First Four, there is a 64 team field. During the first two days of the tournament, 32 games will be played to whittle the field down to 32.

3 Data Collection

3.1 Scores and Spread Data

The sports data needed for this analysis includes the scores and closing spreads of the games of the Round of 64. These scores were obtained through searches of score databases and various news articles covering the spreads at the time. Pre-2011 spread data could not be obtained and is therefore the reason why this investigation only goes back to 2011.

It’s to be noted that this is not a complete set of Round of 64 games from 2011-2026, as some games were both purposefully and not purposefully excluded from this dataset. The 2021 games were not included because, due to the COVID-19 Pandemic, there were limited fans in the arena, largely only people with direct relations to the team attended. A large part of why a location advantage could be significant and not just a random factor with no contextual plausibility to dictate the outcome of the game is because of the fans. If one team has more fans than the other, the support from these fans improves player’s mindsets and therefore makes it closer to a home game environment. The 2021 games do not have this element involved so this study does not take these games into account.

Another set of games that were intentionally excluded are any pushes. Pushes, as described above, are games in which no team “covers the spread.”(if the favorite has a spread of (-6), and they win by 6, it’s a push) Now, if every game had the possibility of being a push, there would be a legitimate 3rd outcome besides “cover” and “don’t cover”. However, most spreads are binary outcomes,

cover vs doesn't cover, which occurs when the spread ends in .5; no team can win a game by a fraction of a point. As such, there is no possibility for a push, and since every game cannot end in a push, this study does not include games in which a push occurred. Note that games in which a push was possible but did not happen are included because there would have been the same binary outcome regardless.

The unintentionally excluded games are games for which there was no available data. These primarily include 1 vs 16 and 6 vs 11 games in the earlier years. Because those matchups were only determined after the First Four games were played, fewer articles were made with the spreads, and fewer records were kept. The articles used from those years primarily cover the 28 games that had both teams determined at the time that the 68-team field was announced.

3.2 Distance/Location Data

When looking at distance, this study decided to use straight-line distance from the college to the city in which the game is held. This because fans may travel to the games in various ways: plane, car, bus, etc. It is not possible to track how every fan is to travel to the game and average the distance out. Therefore, straight-line distance was used to eliminate this variation and reduce bias. This data was collected using Google Maps and Airline Travel Data. Distance is measured in miles.

4 Hypotheses

Three methods were considered to determine if location gives a better edge in spread betting. First, it looks at all games in which one team had to travel more than double the distance of the other. In the attached Python file is the code for this method with the metric used to determine this difference called *DistRatio*. *DistRatio* is equal to the Distance the Lower Seed had to travel divided by the Distance the Higher Seed had to travel. So, the games that fit the double distance traveled requirement are any game with a *DistRatio* value greater than 2 or less than .5. Spreads are set where half the time a team would "cover the spread", and half the time a team would not. So, if location data were to give you an edge on spread betting, you could win over half the time. This allows us to generate the first set of hypotheses:

- **Null Hypothesis:** If a team has to travel less than half of the distance of their opponent, they will cover the spread half of the time.
- **Alternative Hypothesis:** If a team has to travel less than half of the distance of their opponent, they will cover the spread *more* than half of the time.

The second method, instead of looking at the multiplicative difference, looks at the additive difference between the two teams' distance traveled. The stan-

dard deviation of the distance every team traveled across all games was calculated to be 619.88 miles. If any team travels 619.88 miles less than their opponent, it can be considered a significant difference in location. That difference, potentially, could give a better an advantage in picking that team to “cover” the spread. This leads to the second set of hypotheses:

- **Null Hypothesis:** If a team has to travel at least 619.88 miles *less* than their opponent, they will cover the spread half the time.
- **Alternative Hypothesis:** If a team has to travel at least 619.88 miles less than their opponent, they will cover the spread *more* than half the time.

The third method combines the previous two methods; both conditions must be met. A team must travel less than half of the distance of their opponent, and the difference must be greater than 619.88 miles. The hypotheses for this test are as follows:

- **Null Hypothesis :** If a team travels less than half of the distance of their opponent **and** the difference is greater than 619.88 miles, they will cover the spread half the time.
- **Alternative Hypothesis :** If a team travels less than half the distance of their opponent **and** the difference is greater than 619.88 miles, they will cover the spread *more* than half the time.

These are the three tests that the study will conduct, and it will conduct them on seven groups. First, the whole sample will be evaluated, and all games that fit the criterion will be considered. Then, the sample will split into when the Low Seed has the location advantage and when the high-seed has the location advantage. Lastly, the study breaks the seeds into 4 groups and analyzes each. (1-4, 5-8, 9-12, 13-16).

5 Testing and Results

For each of the above hypotheses, the seven groups were each evaluated with a one-sided binomial test. In total, 438 games were collected. The data was analyzed in Pandas, and uses the stats module from Scipy. Here are the results:

Table 1: **Distance Ratio Analysis Results.** n = sample size of games that fit the criteria, x = number of games in which the team with the travel advantage covered the spread, Cover Rate = $100 \cdot x/n$, P-Value = result of one-sided binomial test against $p = 0.5$.

Group	n	x	Cover Rate (%)	P-Value
Whole	208	115	55.288	0.0726
High Seed	68	42	61.765	0.0341
Low Seed	140	73	52.143	0.3364
1-4	91	49	53.846	0.2648
5-8	49	24	48.980	0.5000
9-12	50	29	58.000	0.3222
13-16	18	13	72.222	0.0481

Table 2: **Standard Deviation Distance Analysis Results.** n = sample size of games that fit the criteria, x = number of games in which the team with the travel advantage covered the spread, Cover Rate = $100 \cdot x/n$, P-Value = result of one-sided binomial test against $p = 0.5$.

Group	n	x	Cover Rate (%)	P-Value
Whole	144	76	52.778	0.2799
High Seed	33	56	58.9286	0.1144
Low Seed	88	43	48.8636	0.4576
1-4	53	27	50.943	0.5
5-8	35	16	45.714	0.3679
9-12	42	25	59.524	0.1400
13-16	14	8	57.143	0.3953

Table 3: **Combined DistRatio and Standard Deviation Distance Analysis** n = sample size of games that fit the criteria, x = number of games in which the team with the travel advantage covered the spread, Cover Rate = $100 \cdot x/n$, P-Value = result of one-sided binomial test against $p = 0.5$.

Group	n	x	Cover Rate (%)	P-Value
Whole	118	64	54.237	0.2037
High Seed	44	26	59.091	0.1456
Low Seed	74	38	51.351	0.4538
1-4	47	24	51.064	0.5000
5-8	27	14	51.852	0.5000
9-12	33	19	57.576	0.2434
13-16	11	7	63.636	0.2744

6 Analysis of Results

All results will be interpreted with a significance level of $\alpha = 0.05$

Most tests, including all the tests on the whole dataset, do NOT have $p < 0.05$. Therefore, as a whole, this investigation *cannot* make the general claim that location analysis gives a spread bettor an advantage/edge in the Round of 64 of the Men's March Madness Tournament. Moreover, all tests in the Standard Deviation Analysis did not have $p < 0.05$, meaning there is no evidence to say that additive location differences are something bettors can use to give them an edge.

The multiplicative difference tests, however, do indicate certain advantages. High-seeded teams (9-16) have a 61.765 percent win rate (42/68) with $p = 0.0341$, meaning that there *is* evidence to suggest that in games where the higher-seeded team has a multiplicative location advantage, spread bettors have an edge. Further analysis shows that this advantage primarily comes from 13-16 seeds. So, the investigation concludes that if a higher seed has a multiplicative location advantage, if it's a 13-16 seed, the spread bettor is likely to have an advantage, but if it's a 9-12 seed, proceed with caution and consider other factors.

None of the combined DistRatio and Standard Deviation Analysis difference tests produced a statistically significant result (Test 3). This means, when factoring in the additive difference test to the *already* statistically significant result from the multiplicative difference test, there is no longer a statistically significant result, implying that the advantage comes in smaller additive distance differences. This is plausible, as in smaller additive distance differences from this group, the team with the advantage would be closer to the location of the match when compared to games with bigger additive differences. Looking at the High Seeded and 13-16 results, if the multiplicative distance difference factor was present, but the additive was not, 16/24 High Seeded Teams covered, and 6/7 13-16 Seeded Teams covered. These results are not statistically significant, but suggest that teams with a multiplicative distance advantage that are in close proximity to the city will be where the advantage, if any, lies.

7 Further Study

The most apparent extension of this analysis is to find the spread and scoring data for more games in the Round of 64 in years before 2011, and the games missing from the current data. More data points will be able to better inform bettors on changes over time and if the trends currently observed are a true reflection of reality, or just a byproduct of a small sample size.

8 Sources

- CBS Sports

- FOX Sports
- Sports-Reference
- SB Nation
- Bleacher Report